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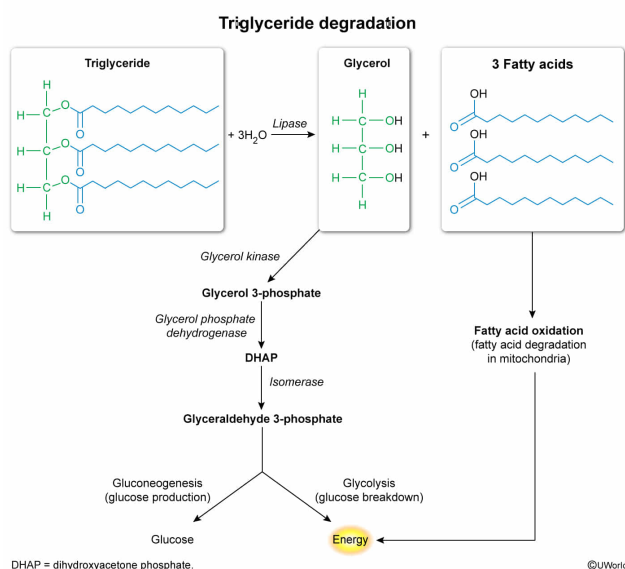
Exogenous dietary lipids, consisting largely of energy storage lipids, are first absorbed by the enterocytes of the small intestine, where they are then packaged into lipoproteins called chylomicrons. In contrast, endogenous energy storage lipids from the liver are packaged into particles known as very-low-density lipoproteins (VLDL). Both chylomicrons and VLDL carry a single apolipoprotein B (ApoB) molecule, which permits their recognition by muscle and adipose tissue; however, in intestinal cells the mRNA for the protein undergoes an RNA editing event that causes the deamination of a cytosine approximately halfway through the transcript. This event results in the intestinal chylomicron isoform of ApoB (ApoB-48) being smaller than the hepatic VLDL isoform (ApoB-100).

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Based on the passage, which of the following lipids are the major lipids deposited by chylomicrons and VLDL into muscle and adipose tissue?

- ☐ A. Sphingolipids [9%]
- ☐ B. Waxes
- ☐ C. Prostaglandins [5%]
- ☒ D. Triacylglycerols [84%]

Explanation:



Lipids are a class of biological macromolecules characterized by their insolubility and **tendency to aggregate** in water. **Lipids have multiple roles** in biochemistry, with the storage of energy being a major function. When energy is needed, these storage lipids can be **catabolized** through various processes to power the production of ATP.

Triacylglycerols are the major example of storage lipids in human biochemistry. Triacylglycerols are composed of **three fatty acids** esterified onto the three hydroxyl groups of a **glycerol** backbone. When energy is needed the fatty acids are **hydrolyzed**, after which they can be catabolized through **β -oxidation** to produce **acetyl-CoA**. The glycerol backbone is phosphorylated, then oxidized, to produce dihydroxyacetone phosphate, which can then continue through to **glycolysis** and the **pyruvate dehydrogenase complex** to produce acetyl-CoA. The acetyl-CoA, produced from either the fatty acids or glycerol, can then go on to enter the **citric acid cycle** and the **electron transport chain** to ultimately produce ATP, which can be used as a source of energy to power various cellular processes.

The passage states that **chylomicrons and VLDLs** are **largely composed of energy storage lipids**. Therefore, these lipid carriers **most likely deposit energy-storing triacylglycerols into muscle tissue**.

(Choice A) **Sphingolipids** are important structural lipids composed of a fatty acid esterified onto a sphingosine molecule. Sphingolipids with a phosphate head group (sphingomyelins) are commonly found as components of the cell membrane. These are generally not used for energy storage.

(Choice B) **Waxes** are fatty acids esterified onto a long-chain alcohol. Although

some organisms catabolize waxes for energy, humans use them mainly for protection and waterproofing (eg, as a major component of cerumen, also known as earwax).

(Choice C) **Prostaglandins** are 20-carbon lipids with a cyclopentane ring. These are important signaling molecules with varied biological effects, but they are not generally used for energy storage.

Educational objective:

Triacylglycerols are a class of lipids that function as energy storage lipids. Triacylglycerols can be catabolized to produce ATP, which can power many cellular processes.

Time spent:0 sec

Subject:
Biochemistry

QID:403440

System:
Biochemistry Lab Techniques

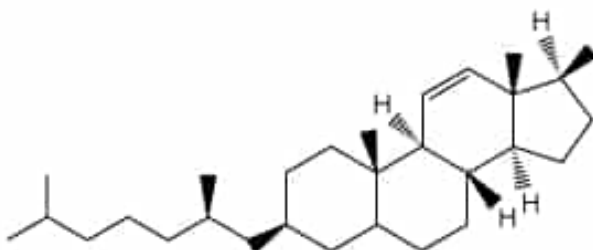
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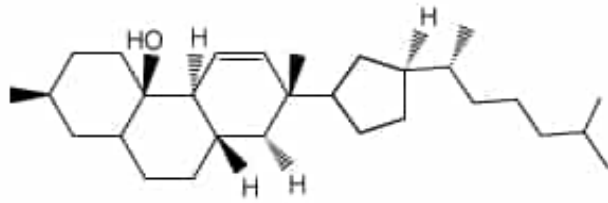
Based on the passage, LDL particles are rich in which of the following molecules?

☐ A.



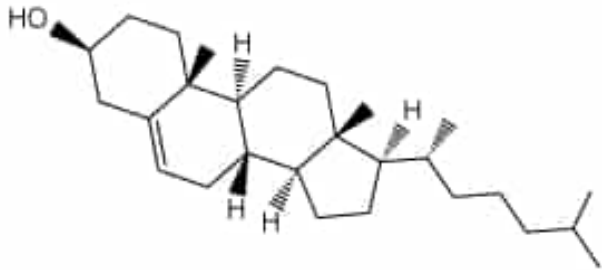
[17%]

☐ B.



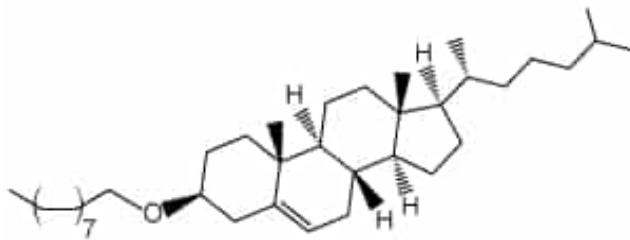
[3%]

☒ C.



[34%]

☐ D.

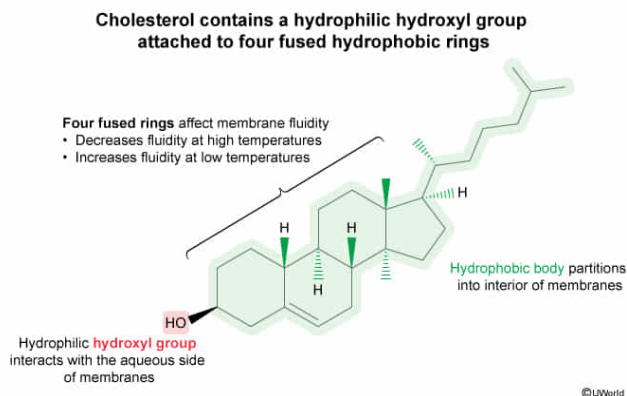


[44%]

Correct

34% answered correctly

Explanation:



The passage states that LDL is enriched in cholesterol, one of the major lipids present in human biochemistry. Cholesterol is not only the precursor molecule from which various downstream molecules are derived, including [steroid hormones](#) and [bile acids](#), but unmodified cholesterol is also a major component of [lipid membranes](#), where it stabilizes the fluidity of the membrane. The roles that

cholesterol plays in the membrane are reflected in its chemical structure.

As the boundary between aqueous compartments, membranes consist of a hydrophilic exterior to interact with the aqueous surroundings and a hydrophobic interior to prevent the passage of polar solutes. Therefore, the components of a bilayer membrane must be **amphiphilic**, comprising both hydrophilic and hydrophobic regions. The hydrophilic functional group on cholesterol is the **hydroxyl group**, which allows cholesterol to interact with the aqueous exterior through **hydrogen bonds**.

As a membrane lipid, cholesterol also stabilizes the **membrane fluidity** by both decreasing fluidity at higher temperatures and increasing fluidity at lower temperatures. Cholesterol is able to accomplish this because its hydrophobic portion is made of **four fused rings**. This fused-ring structure restricts the rotation of the bonds involved, imparting rigidity and *decreasing* fluidity at high temperatures. However, it also impairs packing of the fatty acyl tails of the phospholipid component of the membrane, *increasing* fluidity at low temperatures. Of the choices given, only the molecule shown in **Choice C** contains both the **amphiphilic hydroxyl group** and the **four fused rings**.

(Choice A) This structure lacks the hydroxyl group of cholesterol.

(Choice B) Although this structure has four rings, not all the rings are fused together.

(Choice D) Although the passage states that LDL is enriched in cholesteryl *esters*, this structure shows the hydrocarbon tail attached through an **ether bond** instead of an **ester bond**.

Educational objective:

Cholesterol is a critical component of membranes. Major structural features of cholesterol include the four fused rings and a hydrophilic hydroxyl group.

Time spent:0 sec

Subject:
Biochemistry

QID:403441

System:
Biochemistry Lab Techniques

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After RNA editing of cytosine in the ApoB transcript, what nitrogenous base would the edited structure become?

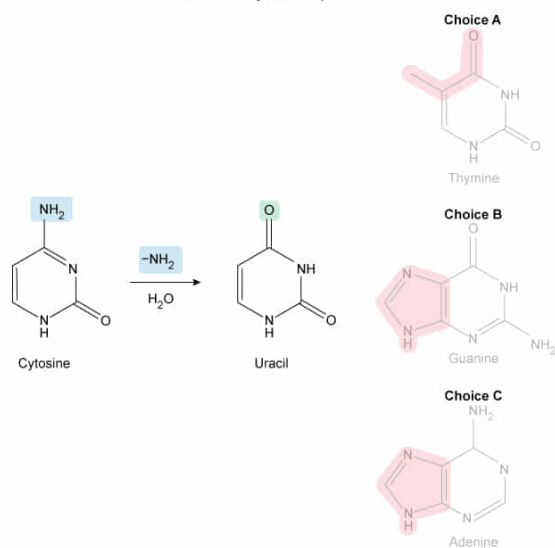
- ☐ A. Thymine [16%]
- ☐ B. Guanine [16%]
- ✓ ☒ C. Uracil [63%]
- ☐ D. Adenine [4%]

Correct

63% answered correctly

Explanation:

Deamination of cytosine produces uracil



The **nitrogenous bases** of **nucleotides** are the chemical groups that encode genetic information within **nucleic acids**. In **replication**, **transcription**, and **translation**, transmission of genetic information relies on the correct pairing of nitrogenous bases on one strand with the nitrogenous bases of a **complementary** strand. These pairs are between one **purine** base and one **pyrimidine** base (guanine pairs with cytosine; adenine pairs with either thymine in DNA or uracil in RNA) through **hydrogen bonds**.

Deamination of nitrogenous bases results in the replacement of an amine group with a carbonyl. Deamination can be either spontaneous or enzyme-catalyzed, and these substitutions can have powerful effects because they change the hydrogen bond characteristics of the base. Deamination reactions result in mutations; for example, **deamination of the pyrimidine cytosine changes it to uracil**.

(Choice A) Although **thymine** (also known as 5-methyluracil) is a pyrimidine that shares the same hydrogen bonding pattern as uracil, it also contains an additional methyl group at position 5 of the heterocyclic ring.

(Choices B and D) Both guanine and adenine are purines and contain an extra fused ring, which would not be created through deamination.

Educational objective:

Deamination changes a single functional group of a nitrogenous base from an amine to a carbonyl, causing mutations that alter hydrogen bonding patterns. The deamination of cytosine results in its conversion to uracil.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403442

System:
Biochemistry Lab Techniques

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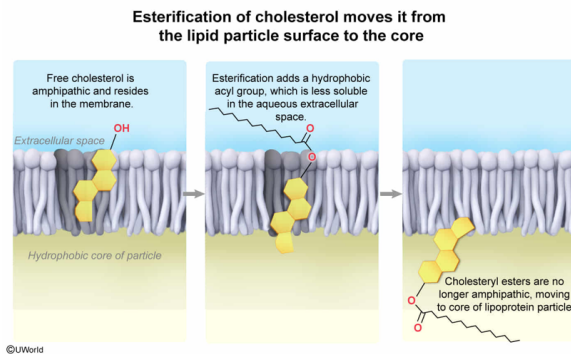
Cholesteryl esters are the result of an esterification reaction between cholesterol and a fatty acid. Which statement best explains the most likely effect of esterification in the lipoprotein?

- ☐ A. The esterification to cholesterol allows the fatty acid to have additional energy storage capacity, creating a more efficient vehicle to transport metabolic fuel across the organism. [40%]
- ☐ B. The acyl group increases the solubility of cholesteryl esters in the blood, allowing them to function as steroid hormones. [14%]
- ☒ C. Esterification results in the transport of cholesterol from the lipoprotein particle surface to the core of the lipoprotein. [24%]
- ☐ D. The addition of the acyl group allows cholesteryl esters to more tightly pack among the phospholipids in the lipoprotein membrane, resulting in a more fluid membrane of the lipoprotein particle. [20%]

Correct

23% answered correctly

Explanation:



Unmodified cholesterol, also known as free cholesterol, is an important component of **lipid membranes**. Cholesterol's place in the membrane results from its **amphipathic** structure—the four hydrophobic fused rings partition into the hydrophobic interior of membranes whereas the hydrophilic hydroxyl group maintains interaction with the aqueous exterior through **hydrogen bonds**.

The esterification of cholesterol would replace the hydrophilic hydroxyl group with a **much less polar ester bond** connected to a large acyl group. This modification reduces the ability of cholesteryl esters to interact with water and destabilizes the localization of cholesteryl esters on the exterior-facing surface of membranes. **Hydrophobic interactions** push the added acyl tail to the interior side of the phospholipid monolayer, and the overall nonpolar cholesteryl ester diffuses from the surface to the core of the lipoprotein particle.

(Choice A) Although the acyl group of cholesteryl esters can be metabolized for energy, no difference exists in energy storage capacity nor gain in efficiency for the acyl group if attached to cholesterol vs. triglycerides, which are also included in lipoprotein particles.

(Choice B) The addition of the hydrophobic acyl group from a fatty acid tail would *decrease* the solubility of a molecule in the blood, which is mostly aqueous.

(Choice D) Tighter packing of the membrane components would result in a *less* fluid membrane, not a more fluid membrane.

Educational objective:

The esterification of cholesterol removes the hydrophilic hydroxyl group and replaces it with a hydrophobic acyl tail. This modification changes the molecule from an amphipathic cholesterol to a hydrophobic cholesteryl ester, resulting in the movement of the ester from the membrane surface to the hydrophobic core of a lipoprotein particle.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403443

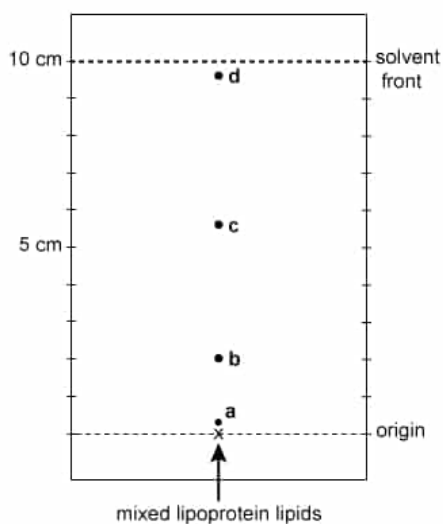
System:
Biochemistry Lab Techniques

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Lipoprotein lipids samples were loaded onto a silica plate and normal-phase thin-layer chromatography was performed using 25% ethyl acetate and 75% hexanes, with the results shown below.



Mass spectrometry confirmed that spots **b** and **d** contain the cholesterol backbone. Based on this information, what is the R_f value of free (nonesterified) cholesterol? (Note: R_f = distance traveled by compound/distance

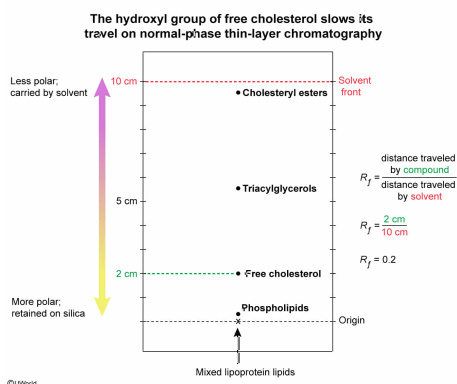
traveled by solvent.)

- ☐ A. 0.025 [9%]
- ✓ ☒ B. 0.20 [39%]
- ☐ C. 0.55 [21%]
- ☐ D. 0.95 [30%]

Correct

39% answered correctly

Explanation:



Chromatography is a technique used to separate molecules based on their different attractions toward the chromatographic matrix (the stationary phase) and the solvent used to carry the molecules (the mobile phase). In **normal-phase thin-layer chromatography**, substances are separated based on **polarity**. **The stationary phase is a polar substance** like silica, and the **mobile phase** is composed of largely nonpolar substances like hexanes and ethyl acetate. Therefore, **polar molecules are retained** to a larger degree on the stationary phase while nonpolar molecules are carried by the solvent as the solvent travels along the matrix.

The question states that both spots **b** and **d** contain the cholesterol backbone, and the passage states that cholesterol and cholesteryl esters are two lipids contained in lipoprotein particles. Unmodified cholesterol contains a polar hydroxyl group whereas in cholesteryl esters, the hydroxyl group is replaced by a nonpolar acyl group attached to cholesterol through an ester bond. Therefore, the nonpolar cholesteryl ester travels farther (spot **d**) and free cholesterol, with an intact hydroxyl group, is more strongly retained by the stationary phase and does not travel as far (spot **b**).

Once the correct spot is identified, the **retention factor** R_f can be calculated using the formula given:

$$R_f = \frac{\text{distance traveled by compound}}{\text{distance traveled by solvent}} = \frac{2 \text{ cm}}{10 \text{ cm}} = 0.20$$

(Choices A and C) These R_f values correspond to spots **a** and **c**, respectively. Spot **a** corresponds to the phospholipids with their highly polar headgroup, and spot **c** corresponds to the nonpolar triacylglycerols.

(Choice D) This R_f value corresponds to spot **d**. Spot **d** traveled farthest and corresponds to the nonpolar cholesteryl esters, not to the amphiphilic free cholesterol.

Educational objective:

Normal-phase chromatography separates molecules based on polarity. Cholesterol is an amphipathic molecule with a hydrophilic hydroxyl group. The esterification of cholesterol removes this polar functional group and makes the molecule nonpolar.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403444

System:
Biochemistry Lab Techniques

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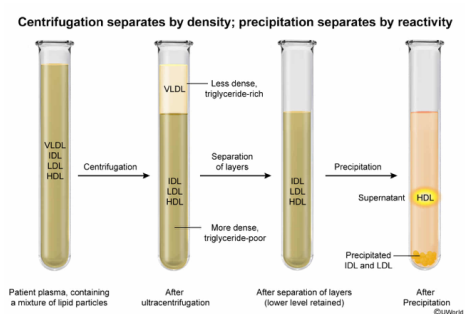
Ultracentrifugation can be used to separate molecules by density and is the gold-standard method for isolating and quantifying plasma lipids. A 5 mL blood plasma sample was spun in an ultracentrifuge, producing two layers: one with storage-lipid-enriched particles and one with storage-lipid-poor particles. The lower layer was treated to precipitate ApoB-containing particles. Which of the following particles remain in the supernatant?

- ✓ ☒ A. HDL only [53%]
☐ B. IDL and LDL only [18%]
☐ C. VLDL only [20%]
☐ D. No lipoproteins remain suspended in the supernatant [7%]

Correct

52% answered correctly

Explanation:



To accurately characterize specific biomolecules, the chemical species of interest must be separated and purified from other interfering contaminants. The techniques involved exploit various physical and chemical

properties of the particles in the mixture to isolate the desired analyte. This question uses centrifugation followed by precipitation in the analysis of lipoprotein particles.

The question states that **centrifugation** is used to separate storage-lipid-rich particles (VLDL) from storage-lipid-poor particles (IDL, LDL, HDL). By quickly spinning samples around a central axis, centrifuges separate particles based on their density: More dense material is forced to the bottom of the sample tube whereas less dense material rises to the top. VLDL's name (very-low-density lipoprotein) indicates that its density is less than that of IDL, LDL, or HDL, so VLDL will rise to the top after centrifugation; in contrast, IDL, LDL, and HDL all sink to the lower layer.

After centrifugation, ApoB-containing particles are precipitated out of the lower layer. The passage states that both IDL and LDL each contain one copy of ApoB, so they would be precipitated by the reaction. HDL alone among the lipoproteins does not contain any isoform of ApoB and would **remain in solution** as part of the **supernatant**.

(Choice B) Both IDL and LDL contain ApoB-100 and would precipitate after reaction with heparin and manganese.

(Choice C) VLDL is rich in storage lipids and would be separated from the other lipoproteins during centrifugation. VLDL's name describes its density as being very low, and so it would settle in the *upper* layer after centrifugation.

(Choice D) HDL *does* remain suspended in the supernatant after precipitation.

Educational objective:

Separation and purification techniques separate substances based on their physical and chemical properties. Centrifugation separates based on density, and precipitation reactions separate based on chemical reactivity.

Time spent:0 sec

Subject:
Biochemistry

QID:403445

System:
Biochemistry Lab Techniques